

1. 🧬 The Multivariate Normal (MVN) Distribution

The Multivariate Normal (MVN) distribution, often denoted $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, describes the joint probability distribution of a vector of random variables, $\mathbf{X} = (X_1, \dots, X_m)^T$, where each variable is itself normally distributed.

A. Building Blocks and Definition

The MVN distribution is defined as any vector whose components are **linear combinations of independent standard normal random variables** ($Z_i \sim N(0, 1)$).

Let $Z_1, \dots, Z_n$ be i.i.d. $N(0, 1)$. The variables $X_1, \dots, X_m$ follow an MVN distribution if they can be expressed as:

$$X_i = a_{i1}Z_1 + a_{i2}Z_2 + \dots + a_{in}Z_n + \mu_i$$

This can be written compactly in matrix notation:

$$\mathbf{X} = \mathbf{AZ} + \boldsymbol{\mu}$$

where $\mathbf{X}$ is an $m \times 1$ vector, $\mathbf{A}$ is an $m \times n$ matrix of constants, $\mathbf{Z}$ is an $n \times 1$ vector of $N(0, 1)$ variables, and $\boldsymbol{\mu}$ is the $m \times 1$ mean vector.

B. Key Properties

- **Closure under Linear Combination:** Any linear combination of an MVN vector is also Normal. If $\mathbf{Y} = \mathbf{CX} + \mathbf{d}$ where $\mathbf{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then $\mathbf{Y}$ is also normally distributed.
- **Marginals are Normal:** Every individual component X_i is, by definition, a **univariate Normal** variable.
- **Parameters:** The MVN distribution is completely characterized by two parameters:
 - i. **Mean Vector ($\boldsymbol{\mu}$):** An $m \times 1$ vector where $\mu_i = E[X_i]$.
 - ii. **Covariance Matrix ($\boldsymbol{\Sigma}$):** An $m \times m$ symmetric, positive semi-definite matrix where the diagonal elements are the variances $\Sigma_{ii} = \text{Var}(X_i)$ and the off-diagonal elements are the covariances $\Sigma_{ij} = \text{Cov}(X_i, X_j)$.

2. 🔥 The Chi-Squared (χ^2) Distribution

The Chi-Squared distribution is a special case of the Gamma distribution and is essential for inference concerning population variance.

A. Formal Definition

If $Z_1, Z_2, \dots, Z_n$ are n **independent Standard Normal** random variables ($Z_i \sim N(0, 1)$), then the random variable Y , defined as the **sum of their squares**, follows a Chi-Squared distribution with n degrees of freedom.

$$Y = Z_1^2 + Z_2^2 + \dots + Z_n^2 = \sum_{i=1}^n Z_i^2$$

$$Y \sim \chi_n^2$$

B. Properties

- **Parameter:** The only parameter is n , the **degrees of freedom (df)**, which equals the number of independent Standard Normal variables being summed.
- **Range:** Since Z_i^2 is always non-negative, the χ^2 distribution is defined only for $y \geq 0$.
- **Shape:** The distribution is highly **skewed** to the right for small n . As $n \rightarrow \infty$, the χ^2 distribution approaches a Normal distribution.
- **Key Statistics:**
 - Mean: $E[Y] = n$
 - Variance: $\text{Var}(Y) = 2n$

3. 🦉 Deriving the Chi-Squared MGF

The derivation of the MGF for the χ_n^2 distribution elegantly uses the MGF superpower for sums and a density function "trick."

Goal: $\phi_Y(t)$ for $Y = \sum_{i=1}^n Z_i^2$

Step 1: MGF Superpower (Product Rule)

Since $Z_1^2, \dots, Z_n^2$ are independent, the MGF of their sum is the product of their individual MGFs:

$$\phi_Y(t) = \phi_{Z_1^2}(t) \cdot \phi_{Z_2^2}(t) \cdot \dots \cdot \phi_{Z_n^2}(t) = [\phi_{Z^2}(t)]^n$$

We only need to find the MGF for a single Z^2 .

Step 2: Finding $\phi_{Z^2}(t)$ using LOTUS

We use the $N(0, 1)$ PDF, $f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$:

$$\phi_{Z^2}(t) = E[e^{tZ^2}] = \int_{-\infty}^{\infty} e^{tz^2} \cdot f_Z(z) dz$$

$$\phi_{Z^2}(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tz^2} e^{-z^2/2} dz$$

Combine the exponents:

$$\phi_{Z^2}(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(z^2\left(t - \frac{1}{2}\right)\right) dz$$

$$\phi_{Z^2}(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(-\frac{z^2}{2}(1 - 2t)\right) dz$$

Step 3: The PDF Integration Trick

We recognize the integral structure as related to a Normal PDF. Recall the general formula for a $N(0, \sigma^2)$ distribution:

$$\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{z^2}{2\sigma^2}\right) dz = 1 \implies \int_{-\infty}^{\infty} \exp\left(-\frac{z^2}{2\sigma^2}\right) dz = \sigma\sqrt{2\pi}$$

By comparison, we set $\frac{1}{2\sigma^2} = \frac{1}{2}(1 - 2t)$.

This gives $\sigma^2 = \frac{1}{1-2t}$, or $\sigma = (1 - 2t)^{-1/2}$.

Substituting $\sigma\sqrt{2\pi}$ for the integral:

$$\phi_{Z^2}(t) = \frac{1}{\sqrt{2\pi}} \cdot (\sigma\sqrt{2\pi}) = \sigma$$

Substitute σ back in terms of t :

$$\phi_{Z^2}(t) = (1 - 2t)^{-1/2}$$

Step 4: Final MGF

Applying the result back to Step 1:

$$\phi_Y(t) = (\phi_{Z^2}(t))^n = \left((1 - 2t)^{-1/2}\right)^n$$

$$\phi_{\chi_n^2}(t) = (1 - 2t)^{-n/2}$$

4. 🔗 The Bridge to Real Statistics (Proposition 2.5)

This proposition is the theoretical foundation for statistical inference involving variance.

Theorem (Distribution of Sample Variance):

If $X_1, \dots, X_n$ are i.i.d. $N(\mu, \sigma^2)$ random variables, and $\bar{X}$ is the sample mean and S^2 is the unbiased sample variance, then:

1. **Independence:** The sample mean ($\bar{X}$) and the sample variance (S^2) are **statistically independent**. (This is a unique property of the Normal distribution.)
2. **Distribution of Scaled Variance:** The statistic formed by scaling the sample variance by its population parameters follows a Chi-Squared distribution with $n - 1$ degrees of freedom:

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$$

Significance of the $n - 1$ Degrees of Freedom

The use of $n - 1$ degrees of freedom is critical. This accounts for the constraint imposed by using the **sample mean** ($\bar{X}$) instead of the true, unknown population mean (μ) in the calculation of S^2 . We lose one degree of freedom because the n deviations ($X_i - \bar{X}$) are not all independent; their sum must equal zero.

This theorem allows statisticians to construct **confidence intervals** for the unknown population variance σ^2 and perform **hypothesis tests** on the variance.

Would you like to review the remaining two distributions built from the Normal: the **Student's t** and the **F-distribution**?